GSA Data Repository # _____

Temperature and Salinity of the Late Cretaceous Western Interior Seaway

Petersen et al.

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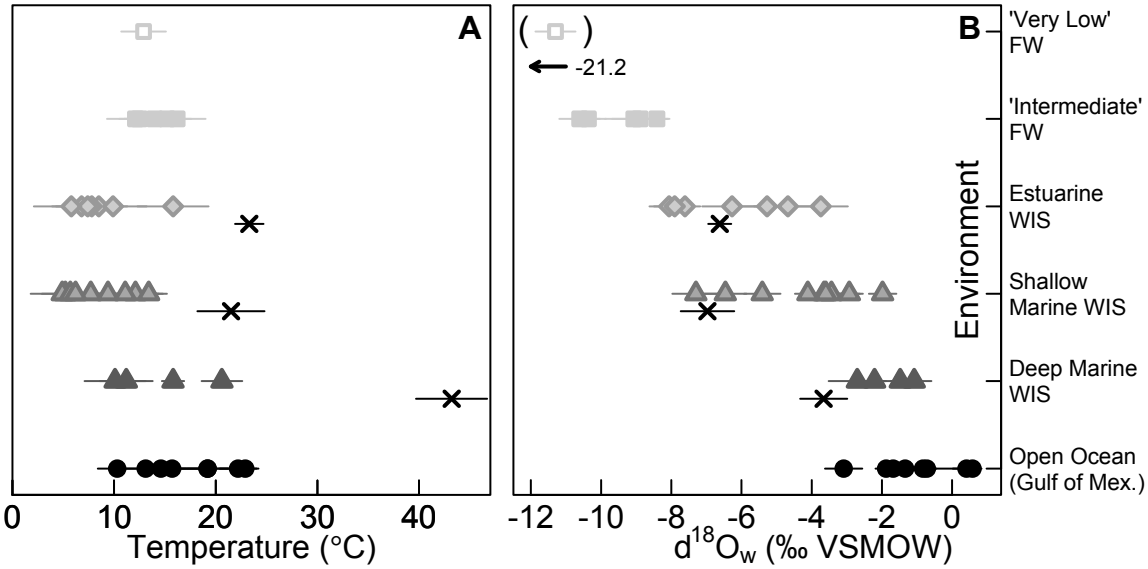
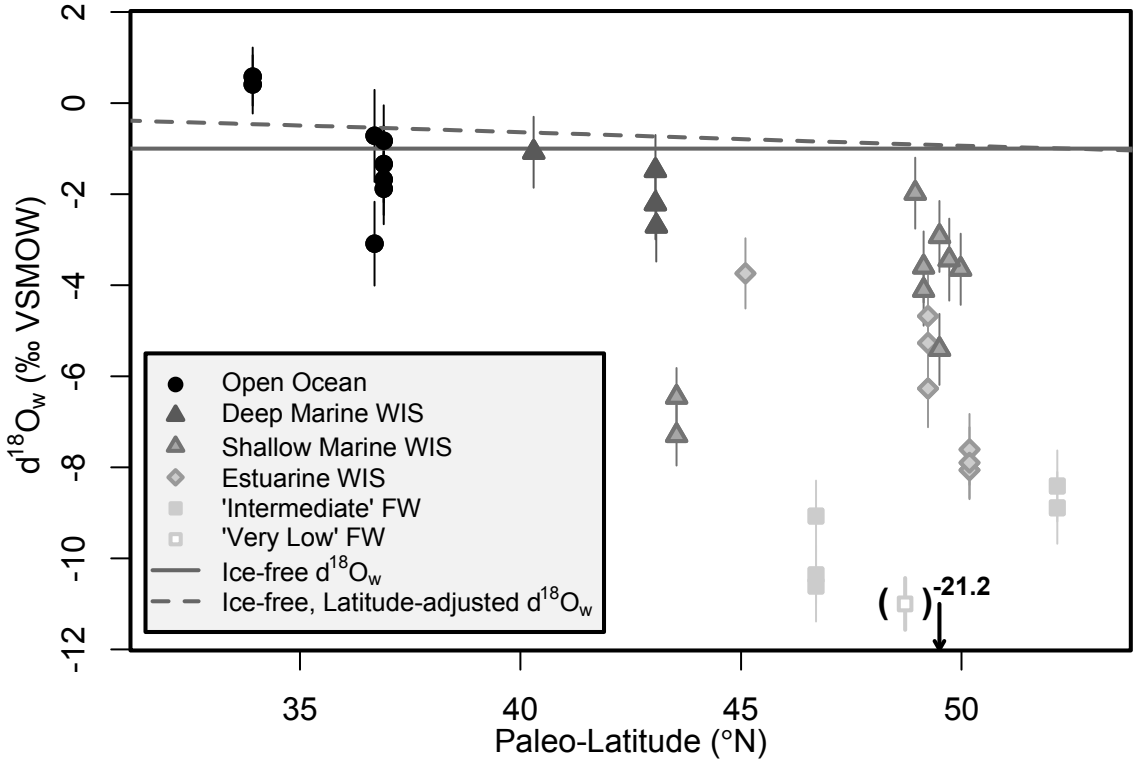
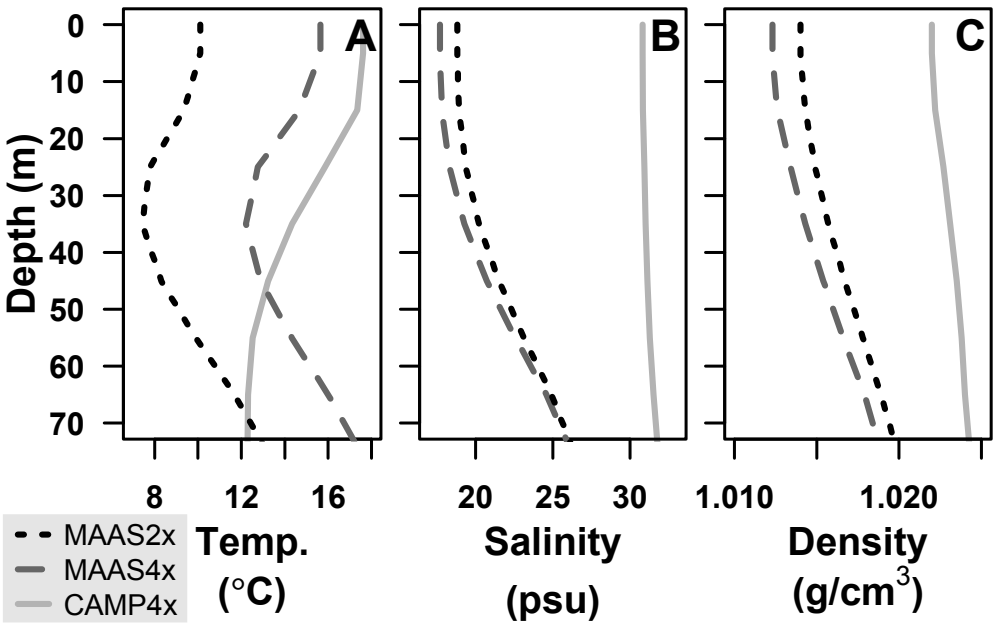


Data Repository Figure 1 (DR1). Comparison of Samples to Co-Occurring Cements
(A) Temperature and (B) $\delta^{18}O_w$ vs. environment for samples and three co-occurring calcite cements (black X's). Cements are offset slightly below samples in each environment category for best visibility. In all cases, temperatures recorded by cements are elevated relative to samples, but within the range expected for shallow burial, and $\delta^{18}O_w$ values are divergent from co-occurring samples. Detailed petrographic and isotopic analysis of shells and diagenetic cements from North Dakota also suggests pristine preservation of shell material in this region, protected by early cementation (Carpenter et al., 1988). $\delta^{18}O$ of cements follow the meteoric calcite line, indicating cements formed in the presence of meteoric waters, likely in a shallow burial setting (Carpenter et al., 1988). Taken together, this suggests samples were not reset or recrystallized during or after early diagenesis when these cements formed and therefore likely record original environmental conditions.

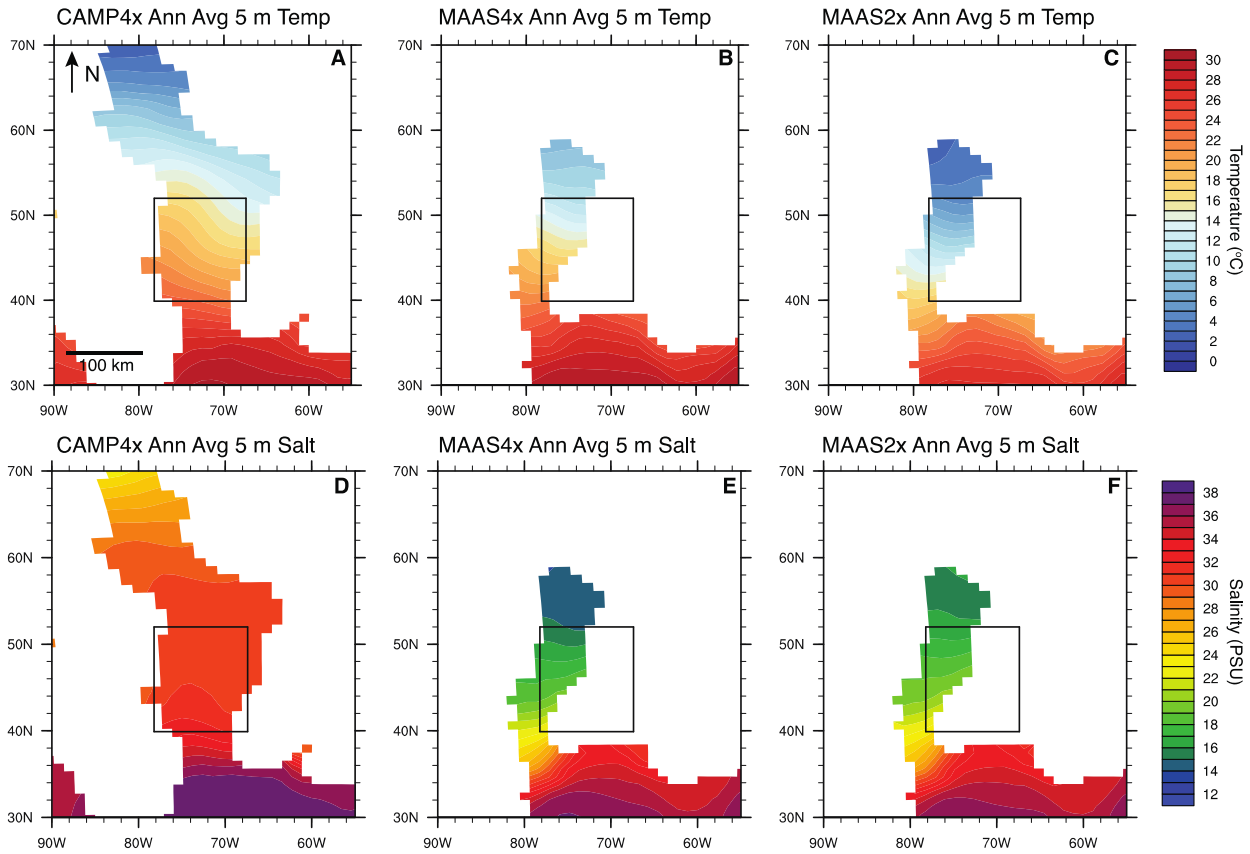


Data Repository Figure 2 (DR2). $\delta^{18}O_w$ vs. Paleo-Latitude

$\delta^{18}O_w$ vs. Paleo-Latitude for samples measured in this study. For comparison, two typical assumptions of $\delta^{18}O_w$ are shown. These are 1) $\delta^{18}O_w$ = -1.0‰ (VSMOW) everywhere (solid line), representing an ice-free world (Shackleton and Kennett, 1975); or 2) the ice-free value adjusted to account for the modern meridional gradient in $\delta^{18}O_w$ (dashed line) (Zachos et al, 1994), giving -0.4 to -1.1‰ (VSMOW) for the latitude range covering the WIS. The data show much more variability than predicted by either of these traditional assumptions, potentially explaining why early attempts at reconstructing WIS paleotemperatures failed to produce reasonable temperatures when they assumed a constant $\delta^{18}O_w$ of -1.0‰ (Tourtelot and Rye, 1969; Wright, 1987; Tsujita and Westermann, 1998; He et al., 2005). Although there appears to be a latitudinal gradient in $\delta^{18}O_w$, we think this is a coincidence based on the latitude at which samples from different environments were collected (most open ocean, deep marine environments from lower latitude relative to estuarine and freshwater environments). Where samples from different environments are found at a similar latitude, the strong relationship shown in main text Fig. 2B is seen clearly (for example, around 48-51°N or around 42-43°N paleolatitude). FW=freshwater, WIS = Western Interior Seaway. Paleo-latitude is in the model coordinate system, taken from Getech Plc paleogeography (Markwick and Valdes, 2004).

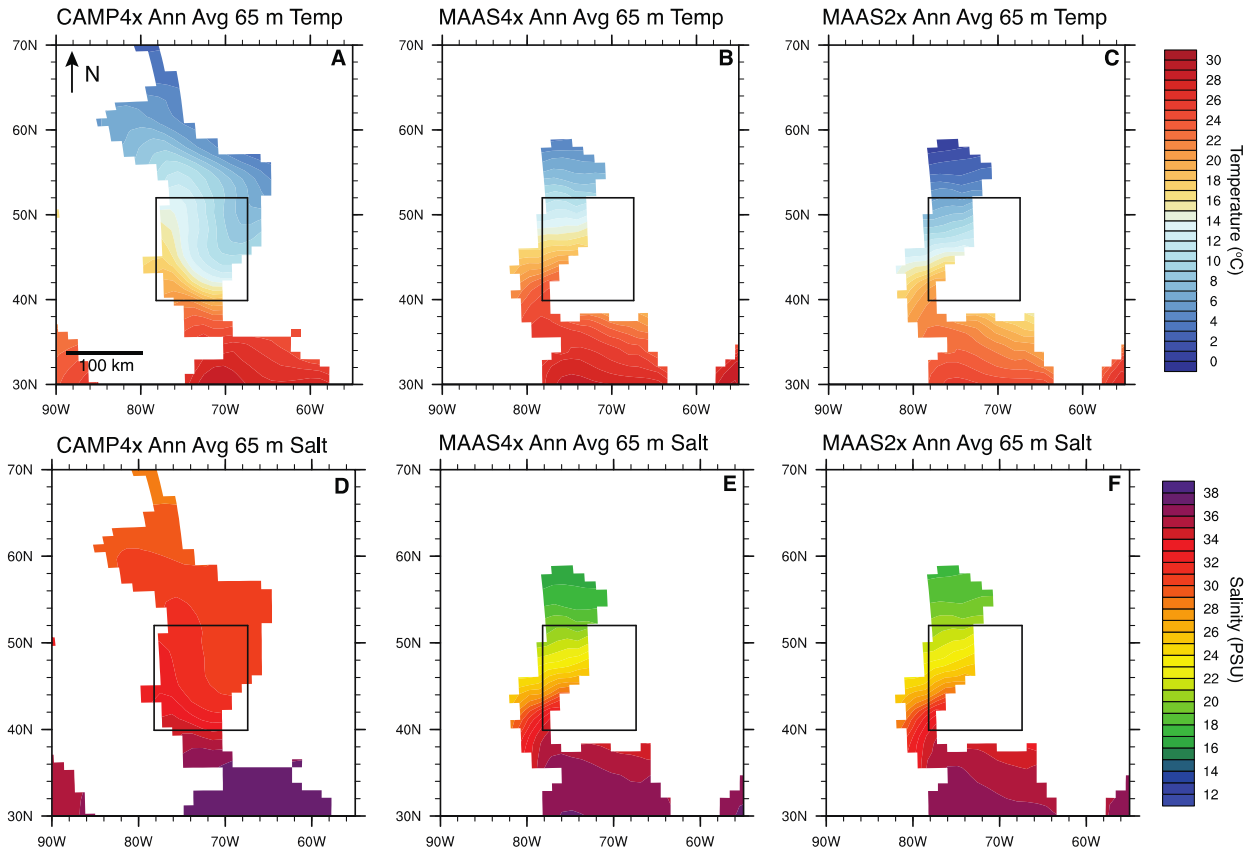


Data Repository Figure 3 (DR3). Depth Profiles of Modeled Seawater Parameters
(A) Temperature, (B) Salinity, and (C) Density depth profiles for three model simulations. Profiles represent annual average conditions, averaged spatially over the sample area (black box in Figs. 1 and 3, DR4, DR5). Similarity between salinity and density profiles for all model scenarios indicates that stratification is salinity-dominated. This suggestion of stratification in the WIS is supported by depth-controlled $\delta^{18}O$ data (Fig. DR11), and previous studies (Erickson, 1974).



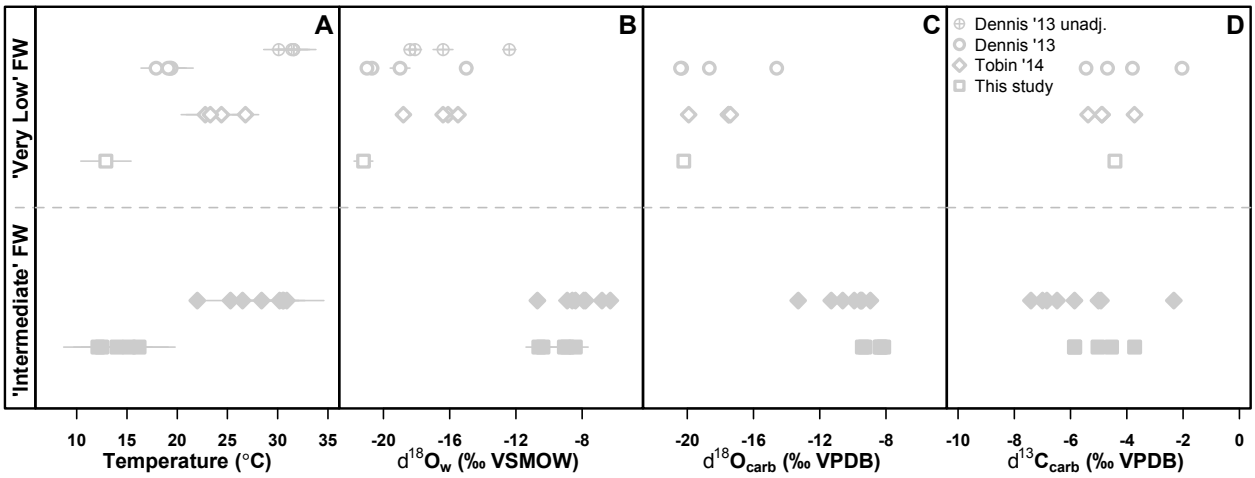
Data Repository Figure 4 (DR4). Modeled Mean Annual Temperature and Salinity at 5-meter depth

CCSM4 model results for three Cretaceous scenarios showing annual average 5-meter temperature (A-C) and 5-meter salinity (D-F), with accompanying color bars. This is the same as Fig. 3, but for 5-meter depth instead of 25-meter depth. Spatial pattern of temperature and salinity are quite similar to 25-meter parameters. Latitude/longitude correspond to Cretaceous paleogeography and are not the same as the modern coordinate frame. Black box denotes sample area shown in Fig. 1.



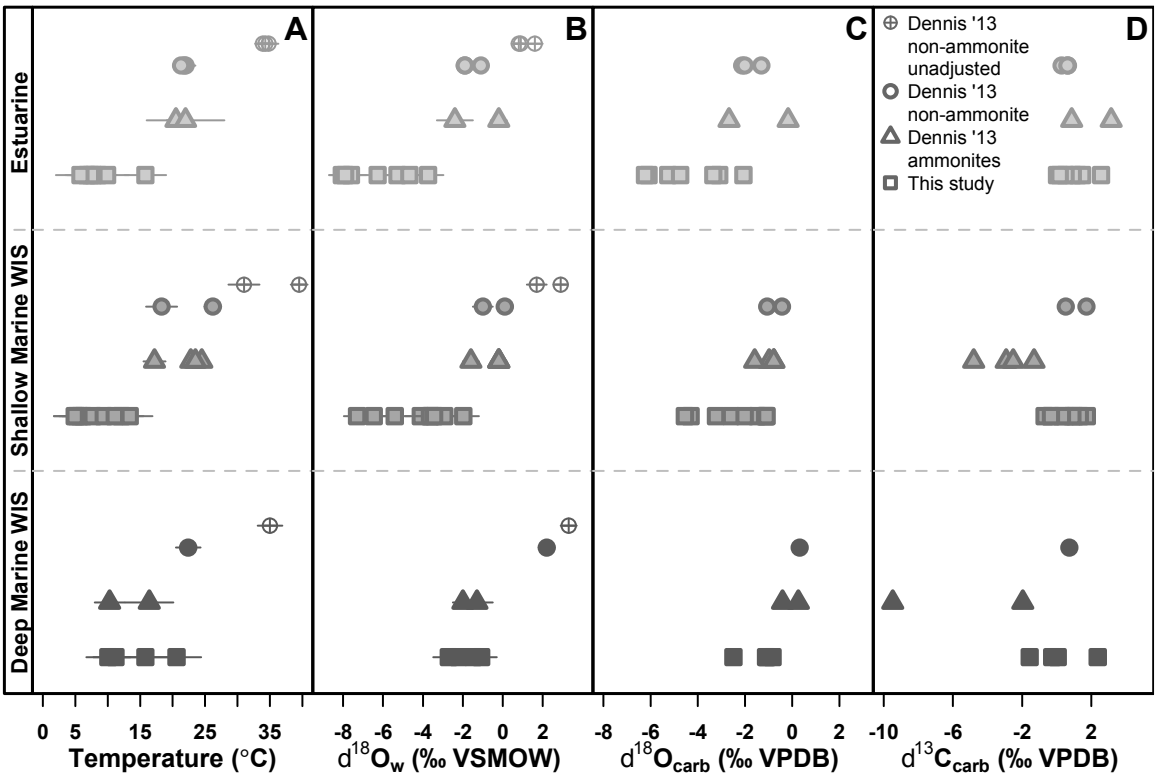
Data Repository Figure 5 (DR5). Modeled Mean Annual Temperature and Salinity at 65-meter depth

CCSM4 model results for three Cretaceous scenarios showing annual average 65-meter temperature (A-C) and 65-meter salinity (D-F), with accompanying color bars. Same as Fig. 3, but for 65-meter depth instead of 25-meter depth. Salinities are increased at depth in the MAAS2x and MAAS4x scenarios relative to 5-meter and 25-meter results (Fig. 3, Fig. DR4). An interesting spatial pattern can be seen in the CAMP4x scenario, showing a warm and salty water mass from the Gulf of Mexico moving up the western edge of the WIS and a cooler and fresher water mass from the northern Arctic region moving down the eastern edge in a gyre-like pattern. This is not so apparent in the 5-meter or 25-meter plots, indicating increased heterogeneity of water masses at depth relative to the surface. Latitude/longitude correspond to Cretaceous paleogeography and are not the same as the modern coordinate frame. Black box denotes sample area in Fig. 1.



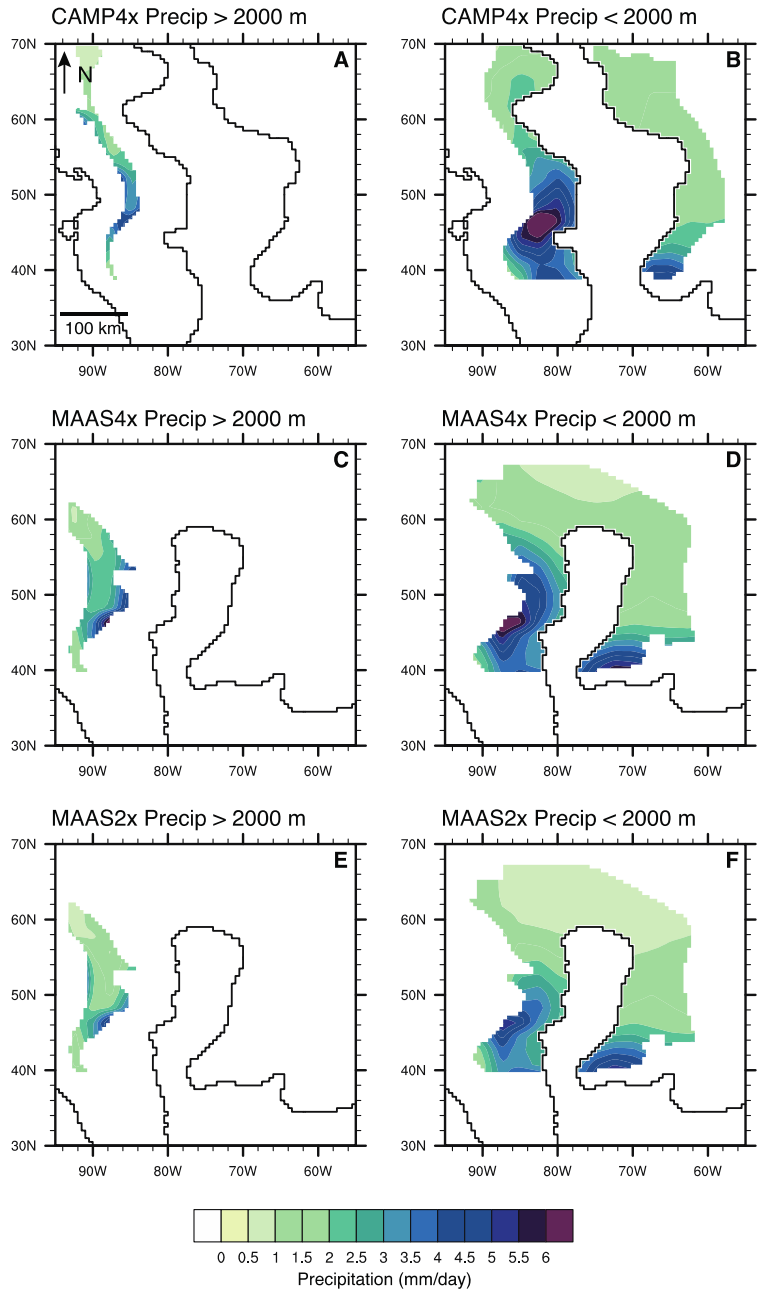
Data Repository Figure 6 (DR6). Comparison to Published Clumped Isotope Data – Freshwater Environments

(A) Temperature, (B) $\delta^{18}O_w$, (C) Carbonate $\delta^{18}O$, and (D) Carbonate $\delta^{13}C$ data for the 'Intermediate' and 'Very Low' freshwater environments from this study compared to data from two previously published studies (Dennis et al., 2013; Tobin et al., 2014). Samples come from the Hell Creek and Lance Formations. Samples from this study include freshwater bivalves (*Unio, Fusconaia, Plesielliptio*) and gastropods (*Campeloma*). Dennis et al. (2013) also measured *Unio*, and Tobin et al. (2014) measured *Unio* and gastropods of unspecified genus. Dennis et al. (2013) incorrectly applied their cephalopod correction to all samples, including bivalves and gastropods, which artificially depressed temperatures for these samples. The open cross circle shows unadjusted bivalve data, which is even warmer and increases the disagreement with the results of this study. Although stable isotope data (C, D) looks similar between all studies, the two previous studies record hotter temperatures, and therefore heavier $\delta^{18}O_w$ values. The large (>10°C) temperature difference between samples from these and our study is unexplained. However, nearby plant-based temperature proxies record a temperature range of 7-17°C for the Hell Creek Formation (Wilf et al., 2003), in line with our measured temperature range and cooler than the two previous studies. For more discussion on the possible reasons for this disagreement, see the "Comparison to Published Clumped Isotope Studies" section of the Data Repository.

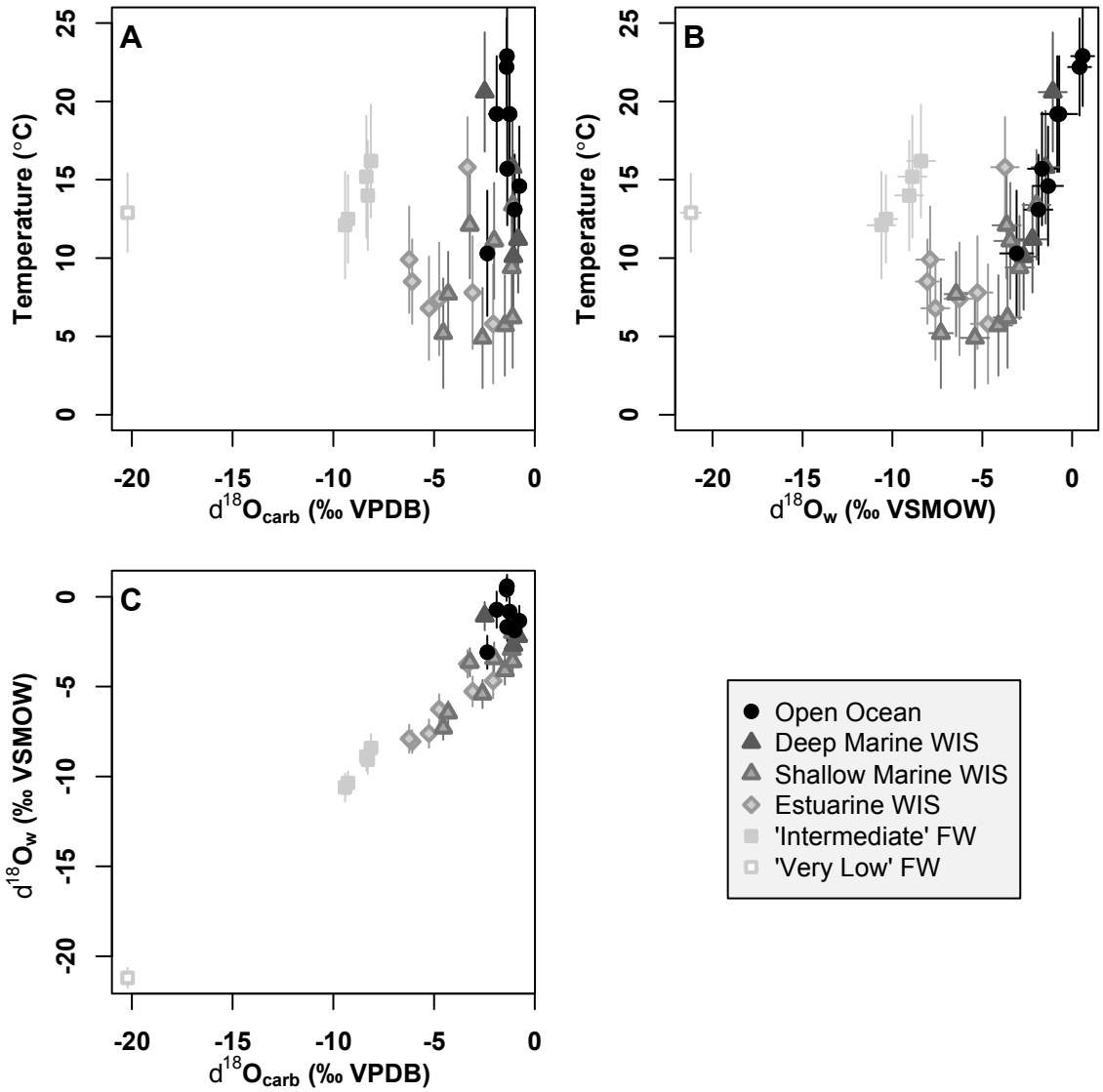


Data Repository Figure 7 (DR7). Comparison to Published Clumped Isotope Data – Marine and Estuarine Environments

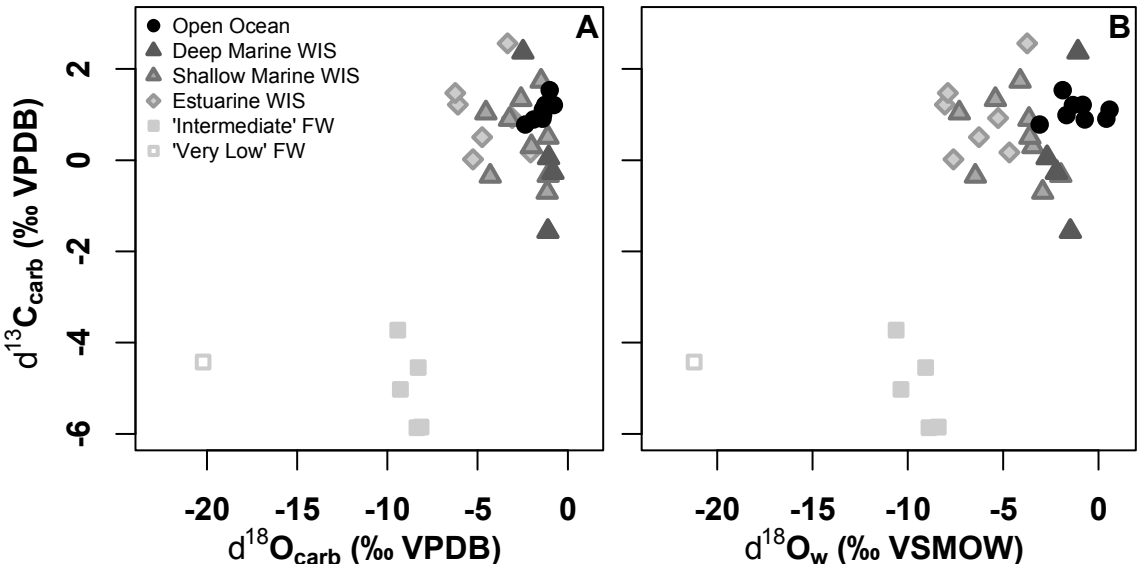
(A) Temperature, (B) $\delta^{18}O_w$, (C) carbonate $\delta^{18}O$, and (D) carbonate $\delta^{13}C$ data for the deep marine, shallow marine, and estuarine WIS environments from this study compared to data from a previously published study (Dennis et al., 2013). Dennis et al. (2013) incorrectly applied their cephalopod correction to all samples, including bivalves and gastropods, which artificially depressed temperatures for these samples (filled circles). The open cross circles show unadjusted bivalve data, which is even warmer and increases the disagreement with results of this study. Ammonites samples are shown in triangles, with the cephalopod correction applied. For comparison between studies, estuarine is said to be the same as "brackish", shallow marine WIS the same as "nearshore interior", and deep marine WIS the same as the "offshore interior" environmental divisions in Dennis et al. (2013). Although stable isotope data (C, D) looks similar (except for ammonites, which sometimes show lighter $\delta^{13}C$ values), data from Dennis et al. (2013) shows hotter temperatures, and therefore heavier $\delta^{18}O_w$ values. This is especially true of $\delta^{18}O_w$ in the estuarine environment, where there is almost no overlap between data from the two studies. Ammonites, but surprisingly also bivalves, do not capture the lighter $\delta^{18}O_w$ values. All samples from Dennis et al. (2013) were collected in South Dakota, near many of our sample locations, so spatial heterogeneity probably does not explain this difference. For more discussion on the possible reasons for this disagreement, see the "Comparison to Published Clumped Isotope Studies" section of the Data Repository.



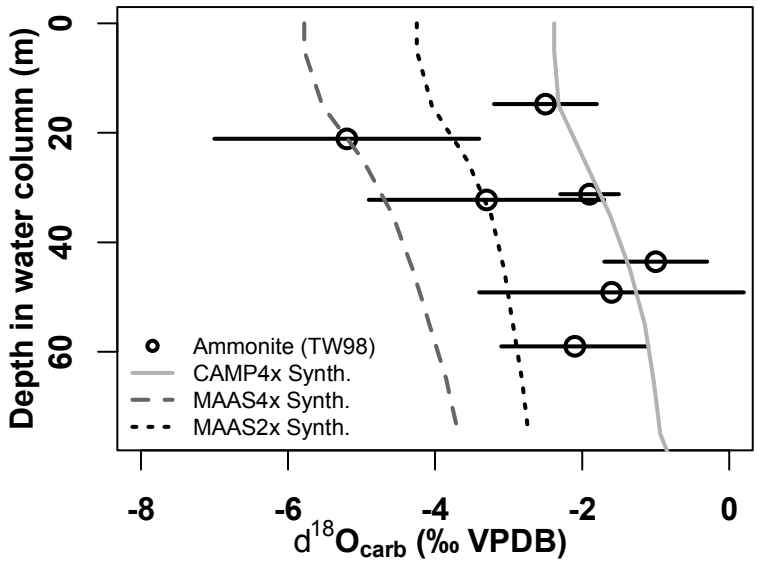
Data Repository Figure 8 (DR8). Total Annual High- and Low-Elevation Precipitation Total Annual Precipitation falling in low- vs. high-elevation in the drainage basin for the WIS, divided at a 2000 m level, for the three model simulations. Combined colored area in low- and high-elevation plots for a given model simulation demonstrates the extent of the drainage basin. Roughly 85% of all precipitation falls below 2000 m elevation. 2000 m is a conservative threshold to divide the low- and high-elevation precipitation regimes. In reality, in order to get precipitation with $\delta^{18}O_w$ values below -20‰, it is likely that higher elevations would be required (~4000 m), meaning an even smaller percentage (<15%) of precipitation would fit in the high-elevation category. The ratio of low- to high-elevation is roughly constant over all simulations, despite variable total precipitation amounts. More precipitation falls in the MAAS4x relative to MAAS2x simulations due to increased temperature and water vapor.



Data Repository Figure 9 (DR9). Correlation between $\delta^{18}O$, Temperature, and $\delta^{18}O_w$
Correlations between (A) Carbonate $\delta^{18}O$ and Temperature, (B) $\delta^{18}O_w$ and Temperature, and (C) Carbonate $\delta^{18}O$ and $\delta^{18}O_w$. By comparing the correlation between carbonate $\delta^{18}O$ and either temperature or $\delta^{18}O_w$, we can directly assess the relative contributions of each to setting carbonate $\delta^{18}O$. We find that in the Maastrichtian WIS, carbonate $\delta^{18}O$ is strongly controlled by $\delta^{18}O_w$ with a much weaker influence by temperature. If temperature were the dominant control, we would expect (A) to show a negative correlation, which it does not. This is one reason why early $\delta^{18}O$-based paleoclimate studies, which assumed an invariant, global mean $\delta^{18}O_w$ values, were unable to produce reasonable temperatures for many WIS environments (Tourtelot and Rye, 1969; Wright, 1987; Tsujita and Westermann, 1998; He et al., 2005). Assuming an invariant $\delta^{18}O_w$ value implies that all changes in $\delta^{18}O$ are due to changes in temperature, whereas we find that most changes in $\delta^{18}O$ are instead due to changes in $\delta^{18}O_w$.

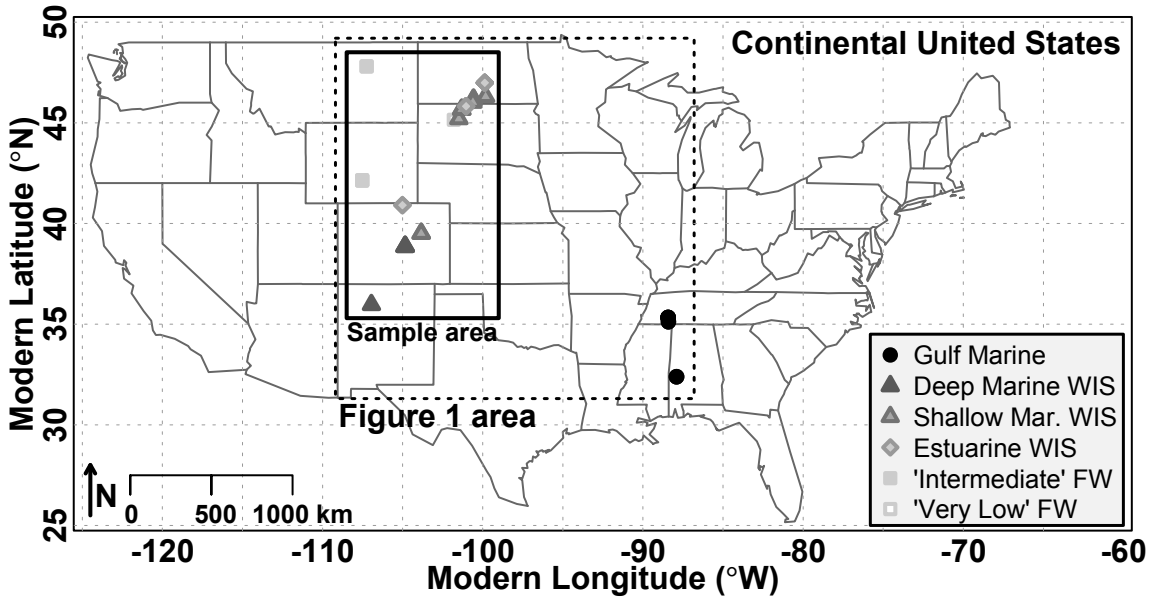


Data Repository Figure 10 (DR10). Carbonate $\delta^{18}O$ and $\delta^{18}O_w$ vs. $\delta^{13}C$
(A) $\delta^{18}O$ and (B) $\delta^{18}O_w$ vs. $\delta^{13}C$, with samples color coded by environment. All freshwater samples are below and all marine samples are above a $\delta^{13}C$ threshold around -3 to -2‰, reflecting the fact that these organisms are sourcing different carbon pools. The fact that $\delta^{13}C$ does not correlate with $\delta^{18}O$ in a linear fashion (i.e. Estuarine environment does not have intermediate $\delta^{13}C$ values, despite having intermediate $\delta^{18}O$ and $\delta^{18}O_w$) indicates that the river and marine waters have different $[HCO^{3-}]$ contents and therefore do not mix linearly, in this case. Specifically, the marine water must have higher $[HCO^{3-}]$ than river water. This is in contrast to linear mixing seen by Carpenter et al. (2003) in *Crassostrea* sp. and *V. vulpes*, which showed intermediate $\delta^{13}C$ and $\delta^{18}O$ values along a linear mixing curve with the 'very low' freshwater end-member. This could reflect variability in $[HCO^{3-}]$ content between trunk rivers and smaller streams, with this study capturing mixing with the tributary streams in our estuarine samples.



Data Repository Figure 11 (DR11). Synthetic carbonate $\delta^{18}O$ depth profiles

Synthetic carbonate $\delta^{18}O$ profiles, compared to measured $\delta^{18}O$ data from ammonites from the Bearpaw Formation, the name for the Late Campanian to Maastrichtian shallow marine units in Canada (Tsujita and Westermann, 1998). Synthetic profiles were calculated by combining the modeled average temperature profile (Fig. DR3A) with the modeled salinity profile (Fig. DR3B) from three model simulations (CAMP4x, MAAS4x, MAAS2x). Salinity is converted to $\delta^{18}O_w$ using the salinity-$\delta^{18}O_w$ relationship determined in the 2-end-member mixing model using the weighted-average freshwater end-member composition. This synthetic salinity was combined with temperature to get $\delta^{18}O$ following the $\delta^{18}O$-Temperature-$\delta^{18}O_w$ relationship for aragonite (Kim et al., 2007). Ammonite data from Tsujita and Westermann (1998), who estimated maximum depth habitat for each ammonite species based on shell morphology. Here we plot mean $\delta^{18}O$ for each species against "average depth habitat" (Max Depth/2). The gradient or range in $\delta^{18}O$ seen between species is similar to the range in $\delta^{18}O$ from surface-to-deep in the synthetic profiles. *P. meeki* (2nd point from the top, ~ 20m depth) is more depleted than other species, but was collected from a more coast-proximal location that was likely more heavily influenced by freshwater (Tsujita and Westermann, 1998). Interestingly, all three synthetic profiles show a similar structure, despite being derived from different temperature and salinity profiles (Fig. DR3). The data best matches the CAMP4x run, which is appropriate, given that the location of the Bearpaw Formation (Canada) is no longer part of the WIS in the Maastrichtian paleogeography.



Data Repository Figure 12 (DR12). Map of entire Continental US showing sample locations Map of entire continental United States, showing locations of samples in a wider geographic context. Symbols denote interpreted paleo-environment. Black solid box outlines 'sample area' over which model results were averaged to determine mean WIS conditions. Black dashed box outlines smaller area shown in Figure 1. Descriptions for all sites can be found below and lat/long coordinates are compiled in Table DR5. WIS = Western Interior Seaway, FW = freshwater.

Data Repository Table DR1. Preservation Indices

Sample Name	Formation (see Locality List for more info)	Aragonite Preserved?	Mother of Pearl Sheen?	Color Band -ing?	Growth Banding?	≥ 1 good pres. indicator?
Good Samples, used in analysis						
A4716-unio	Hell Creek	Y	Y			Y
D5453-campA	Lance	Y				Y
D5453-campB	Lance	Y				Y
D5453-ples	Lance	Y	Y		Y	Y
10022-unioA	Lance	Y	Y			Y
10022-unioB	Lance	Y	Y			Y
16002-ano	Upper Fox Hills*	Y	Y			Y
16002-corbA	Upper Fox Hills*	Y			Y	Y
16002-corbB	Upper Fox Hills*	Y			Y	Y
6113-Luna	Upper Fox Hills*	Y				Y
6113-ost	Upper Fox Hills*	n/a (oyster)			Y	Y
6113-Tamer	Upper Fox Hills*	Y			Y	Y
16216-biv	Upper Fox Hills*	Y				Y
D7904-cymb	Lower Fox Hills*	Y		Y		Y
D7904-tamer	Lower Fox Hills*	Y				Y
A120-1C-cymb	Fox Hills, Timber Lake	Y		Y		Y
A120-C1-cuc	Fox Hills, Timber Lake	Y			Y	Y
SB1-crass	Fox Hills, Timber Lake	n/a (oyster)			Y	Y
A131-Tamer	Fox Hills, Timber Lake	Y			Y	Y
A500-dos	Fox Hills, Timber Lake	Y		Y		Y
A4654-biv	Fox Hills, Trail City	Y	Y, in co-occurring fossils			Y
A4654-gas	Fox Hills, Trail City	Y	Y, in co-occurring fossils			Y
D4153-ost	Lewis Shale	n/a (oyster)				
ChurchAR-biv	Pierre Shale	Y	Y, in co-occurring fossils			Y
BCL-biv	Pierre Shale	Y				Y
Cen1405-blue	Pierre Shale	Y			Y	Y
MC-PRB-EXOa	Prairie Bluff	n/a (oyster)			Y	Y
MC-PRB-EXOb	Prairie Bluff	n/a (oyster)			Y	Y
CC-RIP-CRAa	Ripley	(not meas.)	Y, in co-occurring fossils		Y	Y
CC-RIP-CUCa (umbo-U, ventral margin-V)	Ripley	(not meas.)	Y, in co-occurring fossils		Y	Y
CC-RIP-TURa	Ripley	(not meas.)	Y, in co-occurring fossils		Y	Y
OC-RIP-EXOa	Ripley	n/a (oyster)			Y	Y
OC-RIP-EXOb	Ripley	n/a (oyster)			Y	Y
Samples Thrown Out due to Contamination of Clumped isotope Signal (high Δ_{48})						
9875-ost	Horsethief Sandstone	n/a (calcite)				
D4579-ost	Lewis Shale	n/a (calcite)				
PS1an-inocer	Pierre Shale	Y	Y		Y	Y
ChurchAR-bac	Pierre Shale	(not meas.)	Y			Y
Calcite Cements						
A500-fc	Fox Hills-Timber Lake	n/a (cement)	n/a (cement)	n/a	n/a	n/a (cement)
A668-cem	Hell Creek	n/a (cement)	n/a (cement)	n/a	n/a	n/a (cement)
Cen1504-cem	Pierre Shale	n/a (cement)	n/a (cement)	n/a	n/a	n/a (cement)

*Upper Fox Hills is equivalent to Iron Lightning member. Lower Fox Hills is equivalent to either Timber Lake or Trail City Member. FH = Fox Hills. Y = Yes.

Note: Supplementary Tables 2 and 3 can be found in a separate Excel spreadsheet, where each sub-table can be found in a separate tab and tab names correspond to table names given below.

Data Repository Table DR2. Raw clumped isotope data

Raw data tables containing gas standard, carbonate standard, and sample data from five measurement sessions in (A) January 2015 (B) June 2015 (C) September 2015 (D) December 2015 and (E) February 2016.

Data Repository Table DR3. Sample average clumped isotope data

Mean $\delta^{13}C$, $\delta^{18}O$, Δ_{47}, Temperature, and $\delta^{18}O_w$ values for each sample (one shell, one position) in the clumped isotope data set, calculated as the traditional mean of only 'good' replicates (n=3-5). Replicates flagged as corrupted in Supplementary Table 1 were not included in the mean values. Errors on the mean were taken as 1 s.d. for $\delta^{13}C$ and $\delta^{18}O$ and 1 S.E. for Δ_{47}, Temperature, and $\delta^{18}O_w$ of multiple replicates (internal error). External error for Δ_{47} is taken as the larger of the internal 1 SE or the long-term reproducibility of Carrara standard (1 s.d. = 0.019‰) divided by the square-root of *n* replicates. For two to five replicates, this gives a 1 SE of 0.013‰, 0.011‰, 0.009‰, and 0.008‰. The external error on temperature is calculated as half of T(mean Δ_{47}-extSE) - T(mean Δ_{47}+extSE), where T(*x*) is the Δ_{47}-Temperature calibration function (Defliese et al., 2015) and 'extSE' is the external error replacing the calculated internal SE. The external error on $\delta^{18}O_w$ is assigned to be 0.92‰, 0.78‰, 0.64‰, and 0.57‰ for *n*= 2-5 replicates, respectively, based on typical 1 SE values on $\delta^{18}O_w$ for a sample with a Δ_{47}-error of 0.013‰, 0.011‰, 0.009‰, and 0.008‰. External error is used in all figures.

Data Repository Table DR4. Salinity estimates based on different freshwater end-members

$\delta^{18}O_w$ Value of Freshwater End-Member⇓	**Environment ⇒**	Deep Marine Environment	Shallow Marine Environment	Estuarine Environment
	Range and ***Average*** $\delta^{18}O_w$ values ⇒	-2.7‰ to -1.1‰ ***-1.9‰***	-7.3‰ to -2.0‰ ***-4.3‰***	-8.1‰ to -3.7‰ ***-6.2‰***
-9.5‰ (low-elevation, isotopically 'intermediate' FW composition)		28–35 ***32***	17–31 ***24***	6–24 ***13***
-11.2‰ (85:15 weighted average of 'intermediate' to 'very low' FW compositions)		29–35 ***32***	20–32 ***26***	11–26 ***17***
-21.1‰ (high-elevation, isotopically 'very low' FW composition)		32–35 ***34***	27–33 ***31***	23–30 ***26***

Data Repository Table DR5. Modern and Paleo-Latitude/Longitude Coordinates for sample localities

Locality Identifier	Modern Latitude (°N)	Modern Longitude (°W)	Paleo-Latitude (°N)	Paleo-Longitude (°W)
A4716	45.14	101.82	48.72	70.19
D5453	42.14	107.52	46.70	77.17
10022	47.80	107.25	52.17	74.94
A668	45.49	101.96	48.78	70.33
16002	46.99	99.87	50.18	67.51
6113	45.82	101.04	49.24	69.15
16216	40.91	105.01	45.10	74.90
9875	48.27	112.67	51.83	82.90
D7904	39.51	103.84	43.54	74.09
A120	46.16	100.58	49.5	68.55
SB	46.30	99.82	49.98	68.29
A131	46.02	100.61	49.72	69.19
A500	45.20	101.49	48.95	70.19
A4654	45.67	101.31	49.14	69.48
D4579	42.05	107.44	46.60	77.12
D4153	35.97	106.95	40.30	76.40
PS	43.80	102.52	47.64	71.36
ChurchAR	38.85	104.84	43.06	75.32
BCL	38.85	104.85	43.06	75.33
Cen1504	38.87	104.84	43.08	75.32
MC-PRB	32.39	87.90	33.93	59.76
OC-RIP	35.12	88.41	36.69	59.59
CC-RIP	35.33	88.43	36.90	59.55

**Note: Paleo-Latitude/Longitude correspond to the model coordinates provided by Getech Plc with the paleogeography module.

Supplementary Discussion: Model Description and Configuration

We perform all model experiments with the Community Climate System Model version 4 (CCSM4), which is supported by the National Center for Atmospheric Research (NCAR). CCSM4 performs well in present-day simulations and has been used in previous paleoclimate modeling studies (e.g. Rosenbloom et al., 2013); model details and performance are documented in Gent et al. (2011). Here, we use a fully coupled model configuration that includes: the Community Atmospheric Model version 4 (CAM4), the Community Land Model version 4 with dynamic vegetation (CLM4-DGVM), the Parallel Ocean Project model version 2 (POP2), and the Community Sea Ice model version 4 (CICE4). The atmosphere and land components run on a 1.9x2.5° finite-volume grid while the ocean and sea ice components run on ~1° grid. The atmosphere includes a prognostic aerosol component, which we configure for the Cretaceous using methods similar to Heavens et al. (2012). Further, we adjust the total solar irradiance for the latest Cretaceous based on the calculations of Gough (1981). Orbital configuration and

greenhouse gas concentrations, besides CO_2, are set to pre-industrial values. CO_2 is set to either two times or four times pre-industrial levels (560ppm or 1120ppm), bracketing estimates of Campanian and Maastrichtian atmospheric CO_2 levels from proxies (Nordt et al., 2003). Paleogeographic reconstructions are from Getech, using methods described in Markwick and Valdes (2004). We carry out all simulations for 1500 years, which is ample time for the atmosphere, land, and upper-ocean, including the Western Interior Seaway, to reach near-equilibrium. Presented results are climatologies from the final 30 years of the model simulations. Clumped isotope results are compared with temperature and salinity from 25-meter depth for closest comparison with bottom-dwelling bivalve species, although the conclusions are the same if compared to 5-meter or 65-meter horizons (Fig. DR4, DR5).

Supplementary Discussion: Description of 2-end-member mixing model

We use a simple 2-component linear mixing relationship between two end-member compositions (open ocean and freshwater) to estimate salinity for mixed-water environments (Deep marine WIS, Shallow marine WIS, estuarine WIS). The open ocean end-member is determined by samples from the Gulf of Mexico which have an average $\delta^{18}O_w$ value of -1.1‰ VSMOW, labeled δ_{MAR}, and are assumed to represent fully saline conditions in an ice-free world, set to 34 psu (see below) and labeled S_{MAR}. The freshwater end-member is determined by samples from the Hell Creek and Lance Formations and represents fully fresh waters with a salinity of 0 psu, labeled S_{FR}. The isotopic composition of the freshwater end-member can vary depending on whether you use the low-elevation, isotopically 'intermediate' source (δ_{FR} = -9.5‰), the high-elevation, isotopically 'very low' source (δ_{FR} = -21.1‰), or a weighted average of the two (85:15 average, δ_{FR} = -11.2‰).

With these four inputs, the salinity of any mixed-water environment (S_{mix}) can be estimated based on the water's $\delta^{18}O_w$ value (δ_{mix}).

$$S_{mix} = \left(\frac{\delta_{mix} - \delta_{FR}}{\delta_{MAR} - \delta_{FR}}\right) * (S_{MAR} - S_{FR}) + S_{FR} = \left(\frac{\delta_{mix} - \delta_{FR}}{\delta_{MAR} - \delta_{FR}}\right) * S_{MAR}$$

S_{MAR} represents the salinity of the oceans in an ice-free world. Assuming a modern salinity of 35 psu and a modern ocean volume of 1.33×10^9 km^3 (Charette and Smith, 2010), if the two biggest modern ice sheets melted into the ocean, the ice-free salinity would be:

$$S_{ice-free} = \frac{V_{modern} * S_{modern} + V_{ice\,sheets} * S_{ice\,sheets}}{V_{modern} + V_{ice\,sheets}} = \frac{V_{modern} * S_{modern}}{V_{modern} + V_{ice\,sheets}} = 34.2$$

where $V_{modern} = 1.33 \times 10^9$ km^3, $V_{ice\,sheets} = 2.93 \times 10^6$ km^3 for Greenland (Bamber et al., 2001) plus 2.692×10^7 km^3 for Antarctica (Fretwell et al., 2013), $S_{ice\,sheets} = 0$ psu and $S_{modern} = 35$ psu. The ice-free salinity is calculated to be 34.2 psu, but we round this down to 34 psu based on additional ice found in smaller ice caps (e.g. Iceland) and mountain glaciers not accounted for here.

Supplementary Discussion: Comparison to previous WIS and FW clumped isotope studies

Two previous studies have applied the clumped isotope paleothermometer to mollusks in and around the Western Interior Seaway. Tobin et al. (2014) (hereafter Tobin) looked at freshwater mollusks (*Unionid* bivalves and a few gastropods of unknown genus) from the Hell Creek Formation (Cretaceous) and Fort Union Formation (Paleogene). Dennis et al. (2013) (hereafter Dennis) looked at mainly cephalopods (ammonites, nautiloids) and a few bivalves and gastropods from the Late Campanian to Maastrichtian units of the WIS (Pierre Shale, Fox Hills Fm., Hell Creek), covering the same stratigraphic range studied here.

Observed differences between WIS data from this and the Dennis study:

Fig. DR7 shows a comparison between temperature, $\delta^{18}O_w$, $\delta^{18}O$, and $\delta^{13}C$ data from Dennis and from this study for the three WIS environments (deep marine, shallow marine, estuarine). Generally, the $\delta^{18}O$, and $\delta^{13}C$ data look similar between the two studies. However, two major differences stand out in the temperature and $\delta^{18}O_w$ data. First of all, in our data, the shallow marine and estuarine environments are progressively more depleted in $\delta^{18}O_w$ than the deep marine environment, (-1.9 ± 0.4‰ for deep marine, compared to -4.3 ± 0.6‰ for shallow marine and -6.2 ± 0.6‰ for estuarine). In contrast, in the Dennis study, $\delta^{18}O_w$ values only decrease slightly as the water depth shallows, (-0.4±1.3‰ for offshore interior/deep marine compared to -0.7±0.3‰ for nearshore interior/shallow marine and -1.5±0.4‰ for brackish/estuarine). This difference is most pronounced in the estuarine environment, where Dennis's ammonites and bivalves from this study record $\delta^{18}O_w$ values >5‰ apart. The second major difference is in the temperature data, with bivalve samples from this study producing temperatures of 5-21°C, with all but one between 5-16°C, whereas the Dennis study finds temperatures of 10-26°C, with the majority above 20°C. This 5-10°C difference is seen in all three environments and also in freshwater samples (see below).

Note about cephalopod correction in Dennis study:

The Dennis study contains mainly ammonite samples from the WIS environments, a sample type that had previously not been used with the clumped isotope method. To test the validity of cephalopods as a target for the clumped isotope proxy, the authors performed a modern calibration study on the nearest living relative, *Nautilus*, and found large (>0.05‰, or >10°C) deviations from the established Δ_{47}-temperature relationship seen in other carbonate materials. They then used the average offset (an increase of 0.059‰, Dennis's Table 1) to adjust their paleo-data. However, perhaps erroneously, they also applied this correction to non-ammonite samples in their study (Dennis's Table 3). Figure DR7 shows the effect of this correction on non-ammonite samples, plotting data as published (adjusted for cephalopod offset), and with the cephalopod correction removed (unadjusted). Without this correction, Dennis's bivalve data is even warmer, no longer agrees with the co-occurring ammonite data, and is in greater disagreement with our data.

If the offset in Δ_{47} observed in the calibration study is caused by some kind of cephalopod vital effect, there is no guarantee that the offset calculated for modern *Nautilus* would be constant across species or through time. For example, shallow-water corals, one of the few organisms to show a vital effect in Δ_{47}, produce offsets from the established Δ_{47}-temperature calibration that vary between species and temporally within a single species (Saenger et al., 2012). Additionally, if it is a cephalopod-specific vital effect correction, then it should not have been applied to the non-ammonite samples. It is possible that the apparent offset in *Nautilus* data

was actually an indicator of a transient instrument artifact that was present during the time these samples were analyzed, in which case applying the correction to all samples would be warranted. This is supported by the fact that the ammonites and bivalves from the Dennis study agree when they both have the correction applied, but disagree without the bivalve correction. However, none of the gas or carbonate standards show any indication that something was amiss with the measurement, other than a ~0.010‰ offset in carbonate standards from accepted values (Dennis's Supplementary Table 2).

The Dennis study calculated the cephalopod offset from the Ghosh et al. (2006) Δ_{47}-temperature calibration and then converted the corrected Δ_{47} values into temperature using this same equation. Since this study, it is now accepted that it is best to use a calibration equation defined in the same laboratory in which the unknown samples are analyzed. Repeating this procedure using the calibration of Dennis and Schrag (2010) instead makes very little difference to final temperatures. Our study uses the new, composite Δ_{47}–temperature calibration equation of Defliese et al. (2015), which is very similar to the Dennis and Schrag (2010) equation. Changing calibration equations cannot explain the observed difference.

Possible explanations for differences between WIS data in this and the Dennis study:

We can rule out the latitude at which the samples were collected as a possible explanation for the temperature difference between studies. The Dennis samples come from North and South Dakota, near our northernmost sample site, indicating the Dennis data should be, if anything, colder than our data. We can also probably rule out diagenesis as an explanation. The samples in both studies were rigorously analyzed and found to have sufficient preservation, preserving original aragonite, growth banding, etc. All currently understood methods of diagenesis result in increased Δ_{47}-derived temperatures (other than complete recrystallization under colder conditions, which would be not preserve aragonite). Because the Dennis data is generally warmer than data in this study, this would make it more likely that the Dennis data was altered. However, the Dennis study went so far as to study the mineral fabric under SEM and found pristine crystal textures, making diagenesis of sample material unlikely.

If we take the most harmonious explanation that the "cephalopod-offset" is in fact a necessary correction for all samples and that the Dennis data is correct as published, there are some plausible environmental reasons why the ammonites could be recording different temperature and $\delta^{18}O_w$ values compared to the bottom-dwelling bivalves in our study. Ammonites, unlike sessile bivalves, can move laterally and vertically within the water column. Ammonites could be recording local conditions, but at a different depth in the water column, for example swimming nearer the surface above the bottom-dwelling bivalves. Model simulations (this study) and depth-controlled isotopic studies (Tsujita and Westermann, 1998) suggest that vertical stratification was a pervasive feature of the WIS (Fig. DR3, DR11). Ammonites could also have migrated into estuarine environments from deeper marine waters, or floated after death, such that their shells record open marine WIS temperatures and $\delta^{18}O_w$ values instead of reflecting the environment in which they were found. In this case, care should be taken when trying to reconstruct temperature and $\delta^{18}O_w$ in estuarine and coastal environments, as the ammonites found there may not reflect local conditions.

The higher $\delta^{18}O_w$ values seen in ammonites indicate they inhabited a water mass with salinities closer to open ocean conditions. Combined with water temperatures 5–10°C warmer than those recorded by bivalves, this requires that a warm/salty water mass exist somewhere distinct from the cool/fresh bottom waters in which bivalves were living. Cochran et al. (2003)

suggested that freshwater entered the WIS from below as groundwater. Regardless of the source, the relative position within the water column of the warm/salty ammonite habitat and the cool/fresh estuarine and shallow marine bottom water bivalve habitat must comply with density laws (i.e. must be stable stratification).

Even if ammonites were recording a real and distinct water mass and only moving into the estuarine and shallow marine environments around the time of their death, this still does not explain why the Dennis bivalve data is warmer than bivalve data from this study, with or without the cephalopod/instrument correction. Noticeably, the gradient between $\delta^{18}O_w$ values in different environments is different in the two studies, even if only the non-ammonite samples are considered.

Observed differences in freshwater data between this and the Dennis and Tobin studies:

Fig. DR6 shows a comparison between temperature, $\delta^{18}O_w$, $\delta^{18}O$, and $\delta^{13}C$ data from Tobin, Dennis and this study for the freshwater environment recorded in the Hell Creek and Lance Formations. Generally, the $\delta^{18}O$, and $\delta^{13}C$ data look similar between the three studies, when scatter and the number of samples is taken into account. $\delta^{18}O_w$ data is also similar, but Tobin and Dennis data seem to be slightly higher than data from this study. However, again, the temperatures reconstructed by Dennis are 5–10°C warmer than in this study, and the Tobin temperatures are even higher!

Possible explanations for differences in freshwater data between this and the Dennis and Tobin study:

This large difference in temperature cannot be the result of different taxa measured in each study perhaps recording different environments or seasonal biases. Almost all samples measured in these three studies combined are of the family Unionidae. All Dennis samples are *Unio* sp. (with one unidentified Unionidae). Tobin measured *Unio* sp., with a few gastropods of unspecified genus. With the exception of two samples of the gastropod *Campeloma whitei*, all remaining samples in this study are either *Unio* sp., *Fusconaia brachyopistha* (formerly *Unio brachyopisthus* White), or *Plesielliptio postbiplicatus* (formerly *Unio postbiplicatus*).

This difference can also not be caused by the latitude or location at which samples were collected. The single 'very low' freshwater bivalve measured in this study (A4716-unio, temperature = 12.9±2.2°C) was collected from the same location as samples K6 and K12 from the Dennis study (19.3±1.6°C and 17.9±1.5°C adjusted, ~10°C warmer without the cephalopod correction). One of our 'intermediate' freshwater sites from which two *Unio* bivalves were measured (10022-unioA, temperature = 16.3±1.4°C, 10022-unioB, temperature = 15.2±3.8°C), was only ~40km away from where the majority of Tobin's samples were collected (temperature = 25-30°C). These two facts combined (species, location), mean there is no valid environmental reason why temperatures reconstructed from these three studies should differ, beyond the level of inter-sample variability within a single environment. Additionally, the stable isotope data ($\delta^{18}O$, and $\delta^{13}C$) is quite similar between samples from all three studies, suggesting similar environments for deposition.

One possibility is alteration during sample preparation (frictional heating during drilling, for example), or something similar. This would serve to increase reconstructed temperatures, while not changing carbonate $\delta^{18}O$, leading to heavier calculated $\delta^{18}O_w$ values. Tobin does not specify how sample material was prepared, but this could potentially explain how that study could find much higher temperatures, despite similar carbonate $\delta^{18}O$ values. In this study, some

samples were prepared with mortar and pestle, while others were drilled on the absolute lowest drill speed, only touching the drill bit to the sample for a few seconds at a time in order to minimize the potential for frictional heating. No systematic difference was seen between mortar and pestle and drilled samples.

Dennis specifies that some samples were prepared with a drill and others by 'flaking off portions...using a scalpel'. This makes the resetting by drilling a lower likelihood explanation. Interestingly, the temperature difference between freshwater samples in this study and the Dennis study is similar to the temperature difference seen in the three WIS environments. This suggests some kind of measurement artifact, calibration difference, or laboratory-specific offset that was not accounted for.

In summary, we do not have a good explanation for why these three studies disagree. It does not appear to be due to diagenesis, variations in location of sample collection, or taxa collected (with the exception of ammonites). The differences cannot be eliminated by choosing a different Δ_{47}-temperature calibration. Despite these disagreements, we believe our data makes environmental sense. Our freshwater temperatures agree better with independent plant-based temperature reconstructions (Wilf et al., 2003) and with modeling work. The gradation in $\delta^{18}O_w$ values (and salinities) between the deep marine and estuarine environments makes sense with the interpreted environments and taxa present. Finally, almost all understood ways to alter a measured Δ_{47} value cause an increase in reconstructed temperature and our temperatures are the coldest of the three studies.

Supplementary Discussion: How samples were divided into environments

Many samples used in this study came from museum collections, and were collected by various individuals over many years. The precision of locality descriptions or determination of the formation and unit from which samples were collected varies from collector to collector, and the accepted definitions of the members of the Fox Hills Formation and the boundaries between Hell Creek, Fox Hills, and Pierre Shale have varied somewhat over time (Waage, 1968). We therefore do our best to relate older descriptions to modern ones, and determine environment based on faunal assemblages and other clues in addition to formation. Environmental assignments are described on a site-by-site basis below.

In general, assignments follow this correlation:

Freshwater = Hell Creek and Lance Formations
Estuarine = Fox Hills, Iron Lightning Member and Horsethief Sandstone
Shallow Marine = Fox Hills, Timber Lake and Trail City Members
Deep Marine = Pierre Shale
Lewis Shale = either Deep Marine or Shallow Marine, depending on the location.

We use the term 'Deep Marine' to define the environment in which fine grained shales are deposited, as opposed to 'Shallow Marine' where sands are the predominant texture. This terminology does not indicate a specific water depth, but instead a relative depth. Even at its deepest, the Western Interior Seaway was shallower than typical oceans, reaching only a few 100 meters at its center.

We choose to compare samples by their environment rather than by their age due to the nature of the stratigraphy in the Western Interior Seaway. A single formation may be a different age in a different location, the boundaries between units are sometimes hard to define, and use of

ammonite stratigraphy is no longer possible once the strata are no longer marine in the last few million years of the Maastrichtian (Lance and Hell Creek). This is nicely documented in Figure 3 of Merewether et al. (2011), which shows the variable age and ammonite zone assignments of the studied formations across a wide geographic transect from Wyoming to New Mexico.

Supplementary Information: Detailed Sample Locality Info

Freshwater Environment

A4716

Environment: 'Very Low' Freshwater
Formation: Hell Creek
Samples analyzed from this location (*species name*):

- A4716-unio: *Unio* sp.

Other species found here: n/a
Location: Southeast corner of the SE$^1/_4$ NE$^1/_4$ SW$^1/_4$ sec. 29, T.14N, R.19E. Redelm NE (1951) Quadrangle, Ziebach County, South Dakota, north-northeast-facing bluffs along east side of prairie trail, 2.75 miles south-southeast of Iron Lightning
Collection site description: Basal Hell Creek, in and adjacent to *Unio* bed
Collected by: K. Waage, 1961-1962
Sample obtained from: Scott Carpenter collection
Other identifiers: L6677 (Hartman and Kirkland, 2002), Loc. 304 (Waage, 1968), same location as YPM IP 038033 to 038038, same as K6, K12 (Dennis et al., 2013).

D5453

Environment: 'Intermediate' Freshwater
Formation: Lance
Samples analyzed from this location:

- D5453-campA: *Campeloma whitei* (Russel)
- D5453-campB: *Campeloma whitei* (Russel) (2nd shell)
- D5453-ples: *Plesielliptio postbiplicatus* (Whitfield) (formerly *Unio postbiplicatus*)

Other species found here: *Proparreysia holmesianus* (White), *Tulotomops thompsoni* (White)
Location: South center NW$^1/_4$ NW$^1/_4$ SW$^1/_4$ sec. 28, T. 25N, R. 89W., Elevation 6835 feet, Sta. 547, Carbon County, Wyoming
Collection site description: Beds mapped provisionally as Lance, "Fresh-water mollusks of Lance age"
Collected by: Mitchell W. Reynolds, 1966
Sample obtained from: USGS Denver Collections
Other identifiers: Field locality number F-86 in OF-66-10D

10022

Environment: 'Intermediate' Freshwater
Formation: Lance
Samples analyzed from this location:

- 10022-unioA: *Fusconaia brachyopistha* (White) (formerly *Unio brachyopisthus*)

- 10022-unioB: *Fusconaia brachyopistha* (White) (formerly *Unio brachyopisthus*) (2nd shell)

Other species found here: n/a
Location: Northeast Montana Lignite Oil and Gas field, sec. 32, T. 24N, R. 35E, Valley County, Montana
Collection site description: From horizon of soft sandstone above the massive sandstone above the Bearpaw Fm., Probably Lance
Collected by: H.R. Bennett for A. J. Collier, 1916
Sample obtained from: USGS Denver Collections
Other identifiers: F48

Estuarine Environment

A668

Environment: Estuarine
Formation: Hell Creek
Samples analyzed from this location:

- A668-cem: calcite cement matrix surrounding *Crassostrea subtrigonalis*

Other species found here: *Crassostrea subtrigonalis* (Evans and Shumard), *Corbicula* sp., *Granocardium (Ethmocarbium) aff. G. (E.) whitei* (Dall), *Leptosolen* sp., *Hiatella?* sp. A, *Sphenodiscus lenticularis* (Owen), *Jeletzkytes nebrascensis* (Owen), *Lunatia* sp., *Anomia gryphorhyncha* (Meek)
Location: southeast corner sec. 6, T.14N., R.18E., and northeast corner sec. 7, T.14N., R.18E., Redelm NW (1951) quadrangle, Ziebach County, South Dakota
Collection site description: lowermost Hell Creek, from an oyster bed near the Fox Hills - Hell Creek contact
Collected by: Waage, 1961/1962
Samples obtained from: Scott Carpenter collection
Other identifiers: L6678 (Hartman and Kirkland, 2002), Loc. 75 (Waage, 1968; Speden, 1970), A688 (Carpenter et al., 2003), same site as YPM IP 024103, 024688, 024746, 045828, 045875, 045878, 045892, 046205, 044661, 162401, 162402, 162407-162414, 200285, 237011
Note: Cement matrix surrounds a well preserved *Crassostrea subtrigonalis* (Evans and Shumard) shell, which was not measured for clumped isotopes due to thinness of shell and difficulty extracting enough sample material. It was previously measured for $\delta^{13}C$ and $\delta^{18}O$. $\delta^{18}O$ was between -8 and -10‰ (VPDB) and $\delta^{13}C$ was between -1 and -3‰ (VPDB) (Carpenter et al., 2003). These intermediate $\delta^{18}O$ and $\delta^{13}C$ values fall along a linear mixing line between marine and 'very low' freshwater end-members, making it an estuarine environment.

16002

Environment: Estuarine
Formation: Fox Hills

Samples analyzed from this location:

- 16002-ano: *Anomia micronema* (Meek)
- 16002-corbA: *Corbicula cleburni* (White)
- 16002-corbB: *Corbicula cleburni* (White) (2nd shell)

Other species found here: *Ostrea glabra* (Meek and Hayden), *Neritina loganensis* (Erickson, 1974)

Location: A. Balaban Ranch, ±600 feet west of township line in sec. 36, T. 141N., R. 73W., about 10 miles north of Steele, Kidder County, North Dakota

Collection site description: on south slope of high point on ridge near top of ridge, upper, unnamed, butte-capping sandstone which contains *Crassostrea-Pachymelania* "estuarine" fauna, hillslope above site of abandoned Magnolia Dakota A well, Sibley Buttes

Collected by: R.W. Brown and J. Murata, 1931

Sample obtained from: USGS Denver Collections

Other identifiers: University of North Dakota locality A454 (Erickson, 1974)

Note: Erickson (1974) places this site in the uppermost Fox Hills, perhaps even above the Iron Lightning (later named the Linton Member; Klett and Erickson, 1976), therefore this site was assigned to the estuarine environment, equivalent to the Fox Hills Iron Lightning Member in this study.

6113

Environment: Estuarine

Formation: Fox Hills

Samples analyzed from this location:

- 6113-Luna: *Lunatia subcrassa* (Meek and Hayden)
- 6113-ost: *Ostrea subtrigonalis* (Evans and Shumard)
- 6113-Tamer: *Tancredia americana* (Meek and Hayden)

Other species found here: *Callista* sp.

Location: Standing Rock Indian Reservation, south side of sec. 32, T. 22N., R. 25E., Corson County, South Dakota

Collection site description: Creek bank, near top of Fox Hills

Collected by: A. L. Beekly, 1909 (WR Calvert in charge)

Sample obtained from: USGS Denver Collections

Other identifiers: Original no. 13, 13-09-B

Note: This assemblage contains fauna known to be estuarine. Stanton (1910) describes the fauna in the Standing Rock Indian Reservation and finds this combination of species in the upper Fox Hills Formation. Therefore, this site is assigned to the estuarine environment, equivalent to the Fox Hills Iron Lightning Member due to both the presence of estuarine fauna and the description as "top of Fox Hills". See Calvert et al. (1914) for description of local geology.

16216

Environment: Estuarine

Formation: Fox Hills

Samples analyzed from this location:

- 16216-biv: unidentified bivalve, ~2" wide

Other species found here: n/a
Location: Sec. 21, T. 11N., R. 68W., Larimer County, Colorado
Collection site description: Upper crest Fox Hills sandstone
Collected by: Roy G. Coffin (through J.B. Reeside Jr.), 1932
Sample obtained from: USGS Denver Collections
Other identifiers: 'Alime location where Johson material?'
Note: This site is assigned to be equivalent to the uppermost member of the Fox Hills Formation (Iron Lightning Member) due to description of "upper crest" of Fox Hills.

9875

Environment: Estuarine
Formation: Horsethief sandstone
Samples analyzed from this location:

- 9875-ost: *Ostrea subtrigonalis* (Evans and Shumard)

Other species found here: n/a
Location: Sec. 17, T. 29N., R. 8W., on Dry Fork of Marias River, Pondera County, Montana
Collection site description: Shell bed from top of Horsethief sandstone
Collected by: Eugene Stebinger, 1916
Sample obtained from: USGS Denver Collections
Other identifiers: "5b"
Note: The Horsethief Formation in Montana is equivalent to the Fox Hills Formation and contains both estuarine and shallow marine assemblages (Stebinger, 1915). *Ostrea* is part of the estuarine assemblage. Therefore, this site is assigned to be equivalent to the Fox Hills Iron Lightning Member.

Shallow Marine Environment

D7904

Environment: Shallow Marine
Formation: Fox Hills
Samples collected here:

- D7904-cymb: *Cymbophora warrenana* (Meek and Hayden)
- D7904-tamer: *Tancredia americana* (Meek and Hayden)

Other species found here: *Ethmocardium* sp., *Nucula planomarginata* (Meek and Hayden), *Ptychosyca stantoni* (Henderson), *Euspira obliquata* (Hall and Meek), *Piestochilus* sp.
Location: SW$^1/_4$, sec. 24, T. 6S., R. 58W., Elbert County, Colorado
Collection site description: base of Fox Hills, near contact with Pierre Shale
Collected by: J. A. Sharps, 1971
Sample obtained from: USGS Denver Collections
Other identifiers: "Li18", site marked on the USGS map of Limon Quadrangle (Sharps, 1980)
Note: The description "base of Fox Hills" could indicate the Trail City or Timber Lake Members. This site is likely equivalent to the Timber Lake Member because 1) the *Cymbophora-Tancredia* assemblage is common in the Timber Lake Member (Waage, 1968) and 2) there is a possible hiatus between the Pierre Shale and Fox Hills in central

Colorado (Merewether et al., 2011), meaning the "base" of the FH at this site may be correlative with the middle FH in other places.

A120

Environment: Shallow Marine
Formation: Fox Hills
Samples collected here:

- A120-1C-cymb: *Cymbophora warrenana* (Meek and Hayden)
- A120-1C-cuc: *Cucullaea nebrascensis* (Owen)

Other species found here: *Arctica* cf. *A. ovata*
Location: Sec. 11, T. 130N., R. 79W., Emmons County, North Dakota
Collection site description: Fossils in place, taken from unconsolidated sediment between indurated ledges, Timber Lake Member, Fox Hills Formation, *Cymbophora-Tellinimera* Assemblage Zone (Speden, 1970)
Collected by: J. M. Erickson
Sample obtained from: J. Mark Erickson
Other identifiers: Location described in Erickson (1978) and Carpenter et al. (1988).
Note: Estimated water depth of 18-55 meters (Carpenter et al., 1988), consistent with shallow marine environment.

SB ("Schell Buttes")

Environment: Shallow Marine
Formation: Fox Hills
Samples collected here:

- SB1-crass: *Crassostrea* sp. (likely *Crassostrea glabra*)

Other species found here: *Crassostrea-Pachymelania* biofacies
Location: Schell Buttes, sec. 26, T. 133N., R. 73W., Logan County, North Dakota
Collection site description: A row of buttes composed of hard, dark brown, very fine channel sandstone containing clusters of *Crassostrea glabra* in living position through more than 10 meters of section, upper Fox Hills Fm., Linton Member.
Collected by: J. M. Erickson
Sample obtained from: J. Mark Erickson
Other identifiers: Location described in Erickson (1999).
Note: Formerly a tidal channel with oyster fauna.

A131

Environment: Shallow Marine
Formation: Fox Hills
Samples collected here:

- A131-Tamer: *Tancredia Americana* (Meek and Hayden)

Other species found here: n/a
Location: along Four Mile Creek, eastern Sioux County, North Dakota
Collection site description: Fox Hills Formation, Timber Lake Member
Collected by: J. M. Erickson
Sample obtained from: J. Mark Erickson

Other identifiers: 502, A131 (St. Lawrence University #), described in St. Lawrence honors thesis of Callanan, R.E. (1975).

A500

Environment: Shallow Marine
Formation: Fox Hills
Samples collected here:

- A500-dos: *Dosiniopsis* sp.
- A500-fc: fibrous calcite filling shell

Other species found here: *Nucula scitula*, *Protocardia* sp., *Phelopteria* sp., *Cymbophora warrenana*
Location: Loc. 60 (Waage), Dewey County, South Dakota, roughly 0.5 miles east of SD Rt. 65 and 10 miles north-northwest of Bear Creek Village, north side of Moreau River
Collection site description: 15 feet below the top of the Timber Lake Member, Fox Hills Formation
Collected by: Karl Waage
Sample obtained from: Scott Carpenter collection
Other identifiers: 60-e, Waage Loc. 60, similar to Loc. 59, 61, see Fig. 25 in Waage, 1968.
Note: *Cymbophora-Tellinimera* zone. (also called *Mactra-Tellinimera* zone). A500 assumed to be the same location as A504, but in the Timber Lake Mbr. instead of Trail City Mbr.

A4654

Environment: Shallow Marine
Formation: Fox Hills
Samples collected here:

- A4654-biv: unidentified bivalve (YPM 524505)
- A4654-gas: unidentified spiral gastropod (YPM 524506)

Other species found here: scaphitids, plant material
Location: NW-facing bluff, south side of Grand River, 1.8 miles east-southeast of SD Rt. 65 bridge, and 4.5 miles east-southeast of Black Horse, Corson County, South Dakota
Collection site description: Fox Hills Fm., upper Trail City Mbr., Irish Creek Lithofacies (upper?), middle of fossiliferous part of section
Collected by: K. Waage, J. Reiskind, 1964
Sample obtained from: Yale Peabody Museum via Selena Smith
Other identifiers: IPA. 04654, Catalog #524506, Accn No. 7048, same site as #524507
Note: Two samples have different numbers from the Yale Peabody museum (different concretions), but come from the same locality.

D4579

Environment: Shallow Marine
Formation: Lewis Shale
Samples collected here:

- D4579-ost: *Ostrea glabra* (Meek and Hayden)

Other species found here: *Gryphaea* sp. (?), *Glycimeris wyomingensis* (Meek), *Baculites elliasi, Scaphites (Hoploscaphites)* sp., *Ophiomorpha* sp.

Location: Center S$^1/_2$, NW$^1/_4$, SW$^1/_4$, sec. 15, T. 24N., R. 89W., 500 e/w, 1600 n/s, Carbon County, Wyoming

Collection site description: Lewis Shale, measured section 9, unit 45, 2665-2685 feet above base of Mesaverde Formation, 240 feet (covered interval) above last "Mesaverde" outcrop.

Collected by: M. W. Reynolds, 1964

Sample obtained from: USGS Denver Collections

Other identifiers: OF-64-10D, OF-65-11D (F-35 in this manifest), 4 p. 67. Probably same location as Lot 155 (Fath and Moulton, 1924, p. 28), Separation Rim.

Note: Likely equivalent to Fox Hills in that it underlies the Lance Formation (Fath and Moulton, 1924, Table 1; Reynolds, 1971). The presence of *Glycimeris* indicates a shallow marine environment. In a nearby location, *B. baculus* occurs in the lower part of the Lewis Shale (Reynolds, 1971).

Deep Marine Environment

D4153

Environment: Deep Marine

Formation: Lewis Shale

Samples collected here:

- D4153-ost: *Ostrea patina* (Meek and Hayden)

Other species found here: *Didymoceras* sp., *Oxybeloceras* sp., unidentified ammonite, *Inoceramus* sp., *Fuses* sp. (?), *Ellipsoscapha* cf *subcylindrica* (Meek and Hayden), *Anisomyon* sp., *Trigonia* sp., *Nucula* sp., *Nemodon* sp., *Pecten* sp. (?), *Legumen* sp., *Cymbophora* sp.

Location: Near center of north line, sec. 16, T. 20N., R. 1W., La Ventana quadrangle, Sandoval County, New Mexico

Collection site description: Lewis Shale, approximately 75 feet below top (75 feet below Pictured Cliffs sandstone), position in column is questionable, *B. reduncus* zone, *Didymoceras* sp.

Collected by: E. R. Landis, 1963

Sample obtained from: USGS Denver Collections

Other identifiers: OF-63-7D (L41 in this manifest, same as L40, D4152), OF-64-6D

Note: The Lewis Shale in New Mexico is Campanian in age, equivalent to the Pierre Shale farther north. This site is likely of Middle to Upper Campanian age, based on Cobban et al. (1974).

PS ("Pierre Shale")

Environment: Deep Marine

Formation: Pierre Shale

Samples collected here:

- PS1-inocer: *Inoceramus* sp. (sampled nacreous layer)

Other species found here: n/a

Location: near Scenic, South Dakota, near or in what is now Badlands National Park

Collection site description: Pierre Shale, no further description. Lat/long coordinates chosen as closest outcrop of Pierre Shale (deep marine) to Scenic, SD and the Badlands National Park boundary, which is ~ 2 miles northeast of the town.
Collected by: unkn, 1947
Sample obtained from: Scott Carpenter collection
Other identifiers: from Univeristy of Michigan paleontology collections, defunct

<u>ChurchAR ("Church Across the Road")</u>

Environment: Deep Marine
Formation: Pierre Shale
Samples collected here:

- ChurchAR-biv: unidentified bivalve
- ChurchAR-bac: unidentified baculite

Other species found here: plant material
Location: West Uintah Street, about 90 meters north of intersection with Mesa Road, Colorado Springs, El Paso County, Colorado
Collection site description: Pierre Shale, road cut along north side of West Uintah Street, Invertebrate fossils and concretions in dark grey siltstone matrix.
Collected by: Selena Smith, Kelly Matsunaga, Nathan Sheldon, 2015
Sample obtained from: Selena Smith, Kelly Matsunaga
Other identifiers: Pierre Shale cone-in-cone zone of Lavington (1933), *Baculites baculus-Baculites eliasi* ammonite zone (Scott and Cobban, 1986). Equivalent to D8539, described as Roadcut in NE$^1/_4$ SW $^1/_4$ NE $^1/_4$ sec. 12, T. 14S., R. 67W., El Paso County, Colorado, where *B. eliasi* and *Cymbophora canonensis* were found.

<u>BCL ("Bridge Center Lower")</u>

Environment: Deep Marine
Formation: Pierre Shale
Samples collected here:

- BCL-biv: unidentified bivalve

Other species found here: plant material, unidentified ammonite
Location: Hills behind Colorado Springs Bridge Center (901 N. 17th St.), Colorado Springs, El Paso County, Colorado
Collection site description: Pierre Shale, west-northwest facing slope about 60 meters due south of Bridge Center Building, found in zone containing concretions near top of slope, "upper nodular zone"
Collected by: Selena Smith, Kelly Matsunaga, Nathan Sheldon, 2015
Sample obtained from: Selena Smith, Kelly Matsunaga
Other identifiers: Pierre Shale cone-in-cone zone of Lavington (1933), *Baculites eliasi* ammonite zone (Scott and Cobban, 1986). Near D586, described as 914 19th St., Colorado Springs, NE$^1/_4$ NE$^1/_4$ sec. 11, T. 14S., R. 62W., El Paso County, Colorado, Gregory zone of Pierre Shale, where *B. eliasi* was found.

<u>Cen1504 ("Centennial Blvd.")</u>

Environment: Deep Marine

Formation: Pierre Shale
Samples collected here:

- Cen1504-blue: unidentified bivalve, tinted blue
- Cen1504-cem: crystalline calcite cement, filling cracks

Other species found here: n/a
Location: Centennial Boulevard, south of intersection with West Fillmore Street, Hillside on west side of blocked off road, about 450 meters south of road barricade.
Collection site description: Pierre Shale, shelly bed within a friable shale matrix on southwest-facing slope
Collected by: Selena Smith, Kelly Matsunaga, Nathan Sheldon, 2015
Sample obtained from: Selena Smith, Kelly Matsunaga
Other identifiers: *Exiteloceros jenneyi – Baculites reesidei* ammonite zone (Scott and Cobban, 1986). Near D8537, described as roadcut in NW$^1/_4$ NW$^1/_4$ sec. 2, T. 14S., R. 67W., El Paso County, Colorado, where *Exiteloceros jenneyi* and *Inoceramus vanuxemi* were found.

Open Marine (Gulf Coast)

MC-PRB ("Marengo County")

Environment: Open Ocean
Formation: Prairie Bluff
Samples collected here:

- MC-PRB-EXOa: *Exogyra costata* (Say)
- MC-PRB-EXOb: *Exogyra costata* (Say)

Other species found here: n/a
Location: Alabama Highway 28, 1.6 miles west of Jefferson, Marengo County, Alabama
Collection site description: Roadcuts on both sides of the highway
Collected by: D. B. Macurda Jr., 1967
Sample obtained from: University of Michigan Ruthven Museum of Natural History
Other identifiers: GSA 80th Annual Meeting (Alabama Geological Society), Stop #7, p. 66-67 (Jones, 1967)

OC-RIP ("Owl Creek")

Environment: Open Ocean
Formation: Ripley
Samples collected here:

- OC-RIP-EXOa: *Exogyra costata* (Say)
- OC-RIP-EXOb: *Exogyra costata* (Say)

Other species found here: *Gryphaea* sp.
Location: Owl Creek, McNairy County, Tennessee
Collection site description: unspecified, somewhere along Owl Creek
Collected by: A. R. Cahn, 1938
Sample obtained from: University of Michigan Ruthven Museum of Natural History
Other identifiers: n/a

CC-RIP ("Coon Creek")

Environment: Open Ocean
Formation: Ripley
Samples collected here:

- CC-RIP-CRAa: *Crassitellites vadosus* (Morton)
- CC-RIP-CUCa: *Cucullaea vulgaris* (Morton) ("umbo" position, same shell)
- CC-RIP-CUCa: *Cucullaea vulgaris* (Morton) ("ventral margin" position, same shell)
- CC-RIP-TURa: *Turritella paravertebroides* (Gardner)

Other species found here: *Trigonia* sp., and many others (Wade, 1926)
Location: Dave Week's Farm, Coon Creek, McNairy County, Tennessee. Dave Week's Farm is described as being in the "northeastern part of McNairy County, 3.5 miles south of Enville, 7.5 miles north of Adamsville, and 0.125 miles east of the main Henderson-Adamsville road" by Wade (1926).
Collection site description: unspecified, but "beds containing fossils are best exposed in the valley about 250 yards east of Dave Week's house, along the headwaters of Coon Creek, a small stream flowing northward into White Oak Creek, a tributary of the Tennessee River" according to Wade (1926).
Collected by: I. G. Reimann, 1955
Sample obtained from: University of Michigan Ruthven Museum of Natural History
Other identifiers: n/a

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